

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0045

Winterpock 138 kV
October 20, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0045. The Interconnection Customer proposes a 966.1 MW natural gas combined cycle facility interconnecting at the Winterpock 138 kV substation in the Virginia area. The request was received on June 14, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 40 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$128,843,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Bank Of America to Winterpock 138kV, the Cape Gas Turbine to Winterpock 138kV loads to 551.4% of its 170 MVA rating. Project DFAX: 100.0%.

The Willowbrook to Cape Gas Turbine 138kV line is overloaded to 457.8% of its 199 MVA emergency rating for the loss of the Bank Of America to Winterpock 138kV. The project contributes 100.0% DFAX (966.1 MW) to this overload.

Bank Of America to Winterpock 138kV: 441.5% loading (212 MVA) following the outage of the Cape Gas Turbine to Winterpock 138kV; DFAX = 100.0%.

The Willowbrook Substation to United Cogen Tap 138kV Line is overloaded to 378.6% of its 229 MVA emergency rating for the loss of the Bank Of America to Winterpock 138kV. The project contributes 100.0% DFAX (966.1 MW) to this overload.

For the loss of Bank Of America Substation to Winterpock 138kV line, the Bull Sluice to United Cogen Tap 138kV loads to 348.3% of its 241 MVA rating. Project DFAX: 100.0%.

The Selma to Nottoway Diesel Generating Facility 138kV Line is overloaded to 340.7% of its 224 MVA emergency rating for the loss of the Winterpock to Bank Of America 138kV. The project contributes 86.8% DFAX (838.9 MW) to this overload.

The Selma to Bull Sluice Substation 138kV is overloaded to 327.0% of its 252 MVA emergency rating for the loss of the Winterpock to Bank Of America 138kV. The project contributes 100.0% DFAX (966.1 MW) to this overload.

The Bank Of America to Hull Stree Road 138kV is overloaded to 218.5% of its 246 MVA emergency rating for the loss of the Bank Of America to Dplainville 138kV. The project contributes 58.7% DFAX (567.2 MW) to this overload.

The Seco to Wing Lane 138kV is overloaded to 214.8% of its 167 MVA emergency rating for the loss of the Mitchell Dam to Haggin 345kV. The project contributes 21.3% DFAX (205.6 MW) to this overload.

Harrowgate to Bowmont 138kV: 213.1% loading (231 MVA) following the outage of the Bank Of America Substation to Winterpock 138kV line; DFAX = 37.5%.

The Fruitridge & S. Land Park to Carlin 138kV L is overloaded to 193.0% of its 173 MVA emergency rating for the loss of the Hull Street Road to Bank Of America 138kV. The project contributes 35.4% DFAX (341.8 MW) to this

overload.

The Carlin to Trabue 138kV is overloaded to 191.4% of its 184 MVA emergency rating for the loss of the Hull Street Rd to Bank Of America 138kV L. The project contributes 35.4% DFAX (341.8 MW) to this overload.

For the loss of Mitchell Dam to Haggin 345kV, the Seco to Harvard 138kV loads to 184.1% of its 248 MVA rating. Project DFAX: 21.3%.

The Trabue to Duplainville 138kV Line is overloaded to 180.2% of its 206 MVA emergency rating for the loss of the Bank Of America to Hull St Road 138kV L. The project contributes 35.4% DFAX (341.8 MW) to this overload.

Harrowgate to Yellow Ck 138kV: 177.3% loading (187 MVA) following the outage of the Bank Of America to Winterpock 138kV; DFAX = 25.8%.

For the loss of Hull Street Road to Bank Of America 138kV, the Bank Of America to Duplainville 138kV loads to 176.5% of its 226 MVA rating. Project DFAX: 35.4%.

The Walthall Substation to Haggin 138kV is overloaded to 161.8% of its 236 MVA emergency rating for the loss of the N Tower Tap to Haggin 345kV Line. The project contributes 11.9% DFAX (114.8 MW) to this overload.

For the loss of Mitchell Dam to Haggin 345kV, the Hull Street Road to Harvard 138kV loads to 150.0% of its 333 MVA rating. Project DFAX: 21.3%.

For the loss of Bank Of America to Winterpock 138kV, the Old Hickory to Nottoway Diesel Generating Facility 138kV line loads to 138.8% of its 209 MVA rating. Project DFAX: 35.7%.

Redbed Plains Wind Farm to Fruitridge & S. Land Park 138kV: 135.4% loading (219 MVA) following the outage of the Hull St Road to Bank Of America 138kV; DFAX = 35.4%.

For the loss of Winterpock to Bank Of America 138kV L, the Old Hckory to Bowmont 138kV loads to 134.2% of its 248 MVA rating. Project DFAX: 17.4%.

For the loss of Mitchell Dam to Haggin 345kV, the Wing Lane Substation to Spring St 138kV loads to 128.7% of its 208 MVA rating. Project DFAX: 16.5%.

For the loss of Mitchell Dam to Haggin 345kV, the Elko to Turner Rd 138kV Line loads to 122.5% of its 224 MVA rating. Project DFAX: 6.5%.

The Bank Of America Substation to Bairoa 3008 138kV line is overloaded to 122.4% of its 224 MVA emergency rating for the loss of the Hull St Rd to Bank Of America 138kV. The project contributes 38.9% DFAX (376.1 MW) to this overload.

The Lakeside to Fritch 138kV is overloaded to 121.8% of its 216 MVA emergency rating for the loss of the Mitchell Dam to Skaggs Island Junction 2 345kV. The project contributes 8.5% DFAX (81.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cape Gas Turbine to Winterpock 138kV	Rebuild line with higher-capacity conductor	\$2,983,000	100% (\$2,983,000)
Willowbrook to Cape Gas Turbine 138kV Line	Rebuild line with higher-capacity conductor	\$1,762,000	100% (\$1,762,000)
Bank Of America to Winterpock 138kV	Rebuild line with higher-capacity conductor	\$4,243,000	100% (\$4,243,000)
Willowbrook Substation to United Cogen Tap 138kV Line	Rebuild line with higher-capacity conductor	\$4,843,000	100% (\$4,843,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bull Sluice to United Cogen Tap 138kV	Rebuild line with higher-capacity conductor	\$6,271,000	100% (\$6,271,000)
Selma to Nottoway Diesel Generating Facility 138kV Line	Rebuild line with higher-capacity conductor	\$3,907,000	100% (\$3,907,000)
Selma to Bull Sluice 138kV	Rebuild line with higher-capacity conductor	\$10,699,000	100% (\$10,699,000)
Bank Of America to Hull Stroad 138kV	Rebuild line with higher-capacity conductor	\$1,623,000	100% (\$1,623,000)
Wing Lane to Seco 138kV L	Rebuild line with higher-capacity conductor	\$2,971,000	100% (\$2,971,000)
Harrowgate to Bowmont 138kV	Rebuild line with higher-capacity conductor	\$7,074,000	100% (\$7,074,000)
Carlin to Fruitridge & S. Land Park 138kV	Rebuild line with higher-capacity conductor	\$2,140,000	100% (\$2,140,000)
Carlin to Trabue 138kV	Rebuild line with higher-capacity conductor	\$1,484,000	100% (\$1,484,000)
Seco to Harvard 138kV	Rebuild line with higher-capacity conductor	\$4,437,000	100% (\$4,437,000)
Trabue to Duplainville 138kV Line	Rebuild line with higher-capacity conductor	\$4,305,000	100% (\$4,305,000)
Harrowgate to Yellow Ck 138kV	Rebuild line with higher-capacity conductor	\$2,507,000	100% (\$2,507,000)
Duplainville to Bank Of America 138kV	Rebuild line with higher-capacity conductor	\$1,594,000	100% (\$1,594,000)
Walthal to Haggin 138kV L	Rebuild line with higher-capacity conductor	\$4,873,000	100% (\$4,873,000)
Hull Street Road to Harvard 138kV	Rebuild line with higher-capacity conductor	\$1,473,000	100% (\$1,473,000)
Old Hickory to Nottoway Diesel Generating Facility 138kV	Rebuild line with higher-capacity conductor	\$18,459,000	100% (\$18,459,000)
Redbed Plains Wind Farm to Fruitridge & S. Land Park 138kV L	Rebuild line with higher-capacity conductor	\$12,077,000	100% (\$12,077,000)
Old Hickory to Bowmont 138kV	Rebuild line with higher-capacity conductor	\$17,344,000	100% (\$17,344,000)
Wing Lane to Spring Street 138kV	Rebuild line with higher-capacity conductor	\$1,460,000	100% (\$1,460,000)
Turner Rd to Elko 138kV	Rebuild line with higher-capacity conductor	\$3,195,000	100% (\$3,195,000)
Bank Of America Substation to Bairoa 3008 138kV line	Rebuild line with higher-capacity conductor	\$2,847,000	100% (\$2,847,000)
Lakeside to Fritch 138kV	Rebuild line with higher-capacity conductor	\$4,272,000	100% (\$4,272,000)
	Total allocated to project		\$128,843,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.